

Pratap S

Hosur, Tamilnadu, India

+91 9789122679 • pratapsubramani@gmail.com

pratap834 Pratap S Portfolio

Education

VIT-AP University

Bachelor of Technology in Computer Science and Engineering (AI & ML) – CGPA: 8.46/10

Amaravati, India

2022–2026

Professional Experience

Tanishq

AI Intern

Hosur, India

Jun–Present 2026

- Built an AI system that classifies customer queries by intent, complaint category, department, sentiment, and priority.
- Combined a multilingual XLM-RoBERTa classifier with an LLM fallback, cutting unnecessary LLM calls while keeping accuracy.
- Built a multimodal pipeline (XLM-RoBERTa, Florence, DINOv2, Whisper, OCR, Docling) to process text, image, audio, and document queries.
- Automated the WhatsApp query workflow end-to-end, from intake through AI prediction, routing, and response verification.
- Built full workflow tracking covering routing, escalations, approvals, and automated notifications.

Playlit

Full Stack Developer (Freelance)

Remote

May–Aug 2025

- Architected scalable backend services using REST APIs, Next.js, and Node.js for a music platform serving 400+ users.
- Integrated Clerk Auth for secure authentication and Cloudinary for structured media upload pipelines.
- Worked with PostgreSQL (Neon DB) to manage application data and optimize backend performance.
- Stack: Next.js, Node.js, PostgreSQL (Neon DB), Clerk Auth, Cloudinary.

Projects

AI Job Finder – RAG-Based Matching System:

- Built a retrieval-augmented generation (RAG) system for matching resumes with job listings using semantic search.
- Implemented FAISS vector similarity search using Hugging Face embeddings across 50+ job postings.
- Developed a FastAPI REST API to deliver real-time recommendations and explainable outputs including match scores, skill alignment, and missing requirements.
- Tech Stack: Python, FastAPI, Hugging Face, FAISS.

Real-Time City Emergency Alert & Automation Hub:

- Built a cloud-hosted AI automation hub using n8n to ingest real-time IoT, weather, and citizen-report data for urban emergency detection and response.
- Used Gemini Transformers (NLP) and a Keras image model for incident classification, cutting reporting-to-response time by an estimated 50% via automated WhatsApp/SMS/email alerts.
- Self-hosted on AWS EC2 with a Docker + GitHub Actions CI/CD pipeline, NGINX/Certbot, and CloudWatch-based monitoring (Prometheus, Grafana) with anomaly alerts at >80% CPU.
- Tech Stack: n8n, Docker, GitHub Actions, AWS (EC2, CloudWatch), NGINX, Prometheus, Grafana, Twilio API, Gemini Transformers, Keras.

Quantum-Inspired Portfolio Optimization:

- Formulated stock selection as a Quadratic Unconstrained Binary Optimization (QUBO) problem balancing returns, risk, and portfolio constraints.
- Built a Simulated Annealing solver to search a combinatorial space of stock combinations for near-optimal portfolios.
- Combined quantum-inspired asset selection with classical Sharpe Ratio optimization to compute optimal capital allocation.
- Tech Stack: Python, NumPy, Pandas, SciPy, Simulated Annealing, QUBO.

Technical Skills

Frameworks: PyTorch, Scikit-learn, LangChain, LangGraph, OpenCV, Pandas

Tools: Git, GitHub, Docker, LLM Fine-tuning, AWS, Oracle Cloud, Linux, n8n, OpenClaw, Figma

Languages: Python, Java, R

Certifications

Anthropic's Introduction to Model Context Protocol | AWS Cloud Foundations